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SOLUTION Midterm II Part B

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Dear Students

Attached find solutions for the Midterm II Part B exam.

[EMT_Midterm_II_PartB_Spring_2023_w_Solutions.pdf (https://hulms.instructure.com/courses/2632/files/352355?wrap=1)](https://hulms.instructure.com/courses/2632/files/352355?wrap=1)
(https://hulms.instructure.com/courses/2632/files/352355/download?download_frd=1)

thanks and regards

--sa

Q3 solution by Ailiya:

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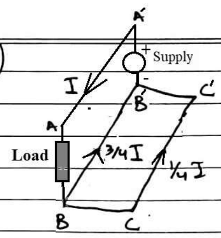
Q3)



Borrowed from the work of
Rizvi, Syeda Ailiya

Some editing is done by the instructor for clarity.

$$\text{dist}(AC) = \sqrt{D^2+D^2} = \sqrt{2}\,D$$

$$L_{total} = L_A + L_B \parallel L_C$$

$$L_A = \lambda_A / I_A$$

$$\lambda_A = 2\times10^{-7} \sum_i I_i \ln(1/D_{Ai})$$

$$= 2\times10^{-7}\left(I_A \ln\left(\frac{1}{D_{AA}}\right) + I_B \ln\left(\frac{1}{D_{AB}}\right) + I_C \ln\left(\frac{1}{D_{AC}}\right)\right)$$

$$= 2\times10^{-7}\left[I \ln\left(\frac{1}{r_A'}\right) + \left(\frac{-3}{4}I\right)\ln\left(\frac{1}{D}\right) + \left(\frac{-1}{4}I\right)\ln\left(\frac{1}{\sqrt{2}D}\right)\right]$$

$$= 2\times10^{-7}\, I \left[\ln\left(\frac{1}{r_A'}\right) - \frac{3}{4}\ln\left(\frac{1}{D}\right) - \frac{1}{4}\ln\left(\frac{1}{\sqrt{2}D}\right)\right]$$

$$= 2\times10^{-7}\, I \left[\ln\left(\frac{1}{r_A'}\right) - \left[\ln\left(\frac{1}{D^{3/4}}\right) + \ln\left(\frac{1}{(\sqrt{2}D)^{1/4}}\right)\right]\right]$$

$$= 2\times10^{-7}\, I \ln\left(\frac{1}{r_A'} \div \frac{1}{D^{3/4}}\cdot\frac{1}{(\sqrt{2}D)^{1/4}}\right)$$

$$= 2\times10^{-7}\, I \ln\left(\frac{D^{3/4}\cdot 2^{1/8} D^{1/4}}{r_A'}\right)$$

$$\boxed{2^{1/2\cdot 1/4} \to 2^{1/8}}$$

$$= 2\times10^{-7}\, I \ln\left(\frac{D\, 2^{1/8}}{r_A'}\right)$$

$$L_A = 2\times10^{-7} \ln\left(\frac{D\, 2^{1/8}}{r_A'}\right)$$

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for $L_B = \dfrac{\lambda_B}{I_B}$

$$\lambda_B = 2\times10^{-7}\left(I_A \ln\left(\frac{1}{D_{BA}}\right) + I_B \ln\left(\frac{1}{D_{BB}}\right) + I_C \ln\left(\frac{1}{D_{BC}}\right)\right]$$

$$= 2\times10^{-7}\, I \left(\ln\left(\frac{1}{D}\right) - \frac{3}{4}\ln\left(\frac{1}{r_B'}\right) - \frac{1}{4}\ln\left(\frac{1}{D}\right)\right]$$

$$= 2\times10^{-7}\, I \left[\frac{3}{4}\ln\left(\frac{1}{D}\right) - \frac{3}{4}\ln\left(\frac{1}{r_B'}\right)\right]$$

$$= \frac{3}{4}\left(2\times10^{-7} I\right)\ln\left(\frac{r_B'}{D}\right)$$

$$L_B = \frac{3\cancel{I}}{\cancel{4}}\left(2\times10^{-7}\right)\ln\left(\frac{r_B'}{D}\right) \Big/ -\cancel{3}/\cancel{4}\cancel{I} = -2\times10^{-7}\ln\left(\frac{r_B'}{D}\right)$$

$$\Rightarrow L_B = 2\times10^{-7}\ln\left(\frac{D}{r_B'}\right)$$

for $L_C = \lambda_C / I_C$

$$\lambda_B = 2\times10^{-7}\left(I_A \ln\left(\frac{1}{D_{CA}}\right) + I_B \ln\left(\frac{1}{D_{CB}}\right) + I_C \ln\left(\frac{1}{D_{CC}}\right)\right]$$

$$= 2\times10^{-7} I \left(\ln\left(\frac{1}{\sqrt{2}D}\right) - \frac{3}{4}\ln\left(\frac{1}{D}\right) - \frac{1}{4}\ln\left(\frac{1}{r_C'}\right)\right)$$

$$\Rightarrow \frac{1}{4}\, 2\times10^{-7} I \left(\ln\left(\frac{1}{(\sqrt{2}D)^4}\right) - \ln\left(\frac{1}{D^3}\right) - \ln\left(\frac{1}{r_C'}\right)\right)$$

$$= \frac{1}{4}\, 2\times10^{-7} I \ln\left(\frac{D^3\cdot r_C'}{2^2 D^4}\right)$$

$$L_C = \frac{1/4\; 2\times10^{-7}\, I \ln\left(r_C'/4D\right)}{-1/4\, I} = -2\times10^{-7}\ln\left(r_C'/4D\right)$$

$$\Rightarrow L_C = 2\times10^{-7}\ln\left(4D/r_C'\right)$$

Q1 Solution by Ammar:

N turns inner radii = a outer radii = b.

(1) height = h Current = I.

(a) For magnetic field intensity with ring.

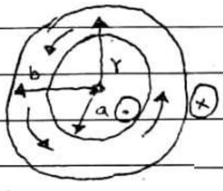
Applying Ampere's law.

$$\oint \bar{H}\cdot d\bar{l} = NI = I_{enclosed} \quad \text{where } d\bar{l} = r\,d\phi\,\hat{\phi}$$

$$H_\phi \int_0^{2\pi} r\,d\phi = NI$$

$$H_\phi (r)(2\pi) = NI$$

$$H_\phi = \frac{NI}{2\pi r}$$



$$\boxed{\bar{H} = \frac{NI}{2\pi r}\hat{\phi}}$$

(b) Since, $\bar{B} = \mu \bar{H}$ let $\mu = \mu_0\mu_r$.

$$\boxed{\bar{B} = \frac{\mu_0\mu_r NI}{2\pi r}\hat{\phi}}$$

Borrowed from the work of Jafri, Syed Muhammad Ammar Ali

(c) $\Phi = \int_S \bar{B}\cdot d\bar{s}$ where $d\bar{s} = dr\,dz\,\hat{\phi}$

$$\bar{\Phi} = \int_{r=a}^{b}\int_{z=0}^{h} \frac{\mu_0\mu_r NI}{2\pi r}\,dr\,dz.$$

$$\bar{\Phi} = \frac{\mu_0\mu_r NI}{2\pi}\int_{r=a}^{b} \frac{z}{r}\Big|_0^h dr$$

$$\bar{\Phi} = \frac{\mu_0\mu_r NI h}{2\pi}\int_{r=a}^{b} \frac{1}{r}\,dr$$

$$\bar{\Phi} = \frac{\mu_0\mu_r NI h}{2\pi}\Big[\ln(r)\Big]_a^b$$

$$\boxed{\bar{\Phi} = \frac{\mu_0\mu_r NI h}{2\pi}\ln\left(\frac{b}{a}\right).}$$

(d)

$$L = \frac{N\bar{\Phi}}{I}$$

$$L = \frac{N\left(\frac{\mu_0\mu_r N \cancel{I} h}{2\pi}\right)\ln\left(\frac{b}{a}\right)}{\cancel{I}}$$

$$L = \frac{\mu_0\mu_r N^2 h \ln(b/a)}{2\pi}\ \text{H}.$$